

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 6 Summer 2025 (Regular)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTCVC602 - FOUNDATION ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

- 1 Which of the following methods is most suitable for obtaining **undisturbed soil samples**?
 - A) Split-spoon sampler
 - B) Open drive sampler
 - C) Standard penetration test (SPT)
 - D) Auger boring
- 2 In which of the following tests is an undisturbed sample mandatory?
 - A) Liquid limit test
 - B) Triaxial shear test
 - C) Specific gravity test
 - D) Sieve analysis
- 3 In the SPT, the split-spoon sampler is driven into the soil using a hammer of weight:
 - A) 50 kg
 - B) 63.5 kg
 - C) 65 kg
 - D) 75 kg
- 4 Which type of failure is characterized by no bulging at the ground surface and sudden punching of the footing into the soil?
 - A) General shear failure
 - B) Local shear failure
 - C) Punching shear failure
 - D) Lateral shear failure
- 5 Safe bearing capacity is defined as:
 - A) Load causing elastic deformation
 - B) Load at which failure occurs
 - C) Ultimate capacity divided by factor of safety
 - D) Pressure from wind loads
- 6 Terzaghi's bearing capacity equation is applicable for:
 - A) Deep foundations
 - B) Cohesive soils only
 - C) Shallow strip footings
 - D) Raft foundations

- 7 Which type of footing is recommended when the soil bearing capacity is very low?
 A) Isolated footing
 B) Combined footing
 C) Raft (Mat) footing
 D) Trench footing
- 8 A common clay mineral responsible for swelling behavior in expansive soils is:
 A) Kaolinite
 B) Illite
 C) Montmorillonite
 D) Chlorite
- 9 Negative skin friction occurs when:
 A) Ground settles faster than the pile
 B) Pile settles faster than the soil
 C) Soil is fully consolidated
 D) Pile is too short
- 10 Sheet piles are primarily used to:
 A) Increase bearing capacity
 B) Provide vertical support for buildings
 C) Retain soil and resist lateral pressure
 D) Support bridge decks
- 11 The Factor of Safety (FOS) in slope stability is defined as:
 A) Ratio of driving forces to resisting forces
 B) Ratio of resisting forces to driving forces
 C) Difference between resisting and driving forces
 D) Sum of driving and resisting forces
- The stability number in slope stability refers to:
 A) A dimensionless number indicating slope stability for given soil parameters
 B) Soil moisture content
 C) Slope angle in degrees
 D) Total load on the slope
- Q2. Solve the following.
- A) What are the purposes of soil exploration? 6
- B) Explain the Electric Resistivity Method? 6
- Q3. Solve the following.
- A) State and explain the below terms 6
 1) Net Pressure Intensity,
 2) Net Ultimate bearing capacity, and
 3) Net Safe bearing capacity
- B) Explain the terms in Terzaghi's bearing capacity eq. developed using general shear failure? 6
- Q4. Solve Any Two of the following.
- A) What are expansive soils? Explain in detail the characteristics of expansive soils? 6
- B) Define and explain the following terms: 6
 1) Free Swell
 2) Swelling Potential
 3) Swelling Pressure
- C) Define settlement of foundation. Explain the different types of settlement in shallow foundations. 6
- Q5. Solve Any Two of the following.

- A) A group of 9 piles, with 3 piles in a row, was driven into soft clay extending from ground level to a great depth. The diameter and length of piles were 30 cm and 10 m, respectively. The unconfined compressive strength of clay is 70 kN/m^2 . If the piles were spaced at 90 cm c/c , compute the allowable load on the pile group on the basis of shear failure criteria for a factor of safety of 2.5. Neglect bearing at the tip of the piles. Take $m = 0.6$ for shear mobilization around each pile. 6
- B) Explain the sheet pile walls in detail using neat sketches. 6
- C) State and explain in detail the Engineering News Record Formula (ENR) and Hiley's Formula? 6

Q6. Solve Any Two of the following.

- A) What are the different types of slopes. Explain in detail with neat sketch and example? 6
- B) A slope is to be constructed at an inclination of 30° with the horizontal. Determine the safe height of the slope at factor of safety of 1.5. The soil has the following properties: $c = 15 \text{ kN/m}^2$, $\phi = 22.5^\circ$ and $\gamma = 19 \text{ kN/m}^3$ 6
- C) Enlist and explain in detail the different types of slope failures? 6

*** End ***